

# ASSIGNMENT - 1

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# Code :

```
// it is single inheritance program
class f{      // create a class
    int a =10;
    void sum(){
        System.out.println("sum is " + (a+34));
in class f have initialise variable and one method
    }
}

    // create another class
// class f is inherited by class S to using of keyword " extends"
/* here class f is super class
and class S is sub class*/
    class s extends f{
        int b = 54;
        // f add = new f();
        void sub(){
            System.out.println( " subtraction is " + (b-10) );
        }

    }

    public class Inherit1 { // main method
    public static void main(String[] args) {
        // create an object for sub class
        s oc = new s();
        // print and call super class variable
        System.out.println(" variable can access is" + (oc.a) );
        // print the sub class variable
        System.out.println(" variable can access" + (oc.b));
        // call super class methods from sub class object
        oc.sum();
        oc.sub();

    }
}
```

result :

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Inherit1.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Inherit1
variable can access is10
can access54
sum is 44
subtraction is 44
```



# Second type :

// it is second type of inheritance is multi inheritance

create three class one class is inherited another class and another class is inherited by other class

```
class animal{
    String name = " lion";
    void ani(){
        System.out.println(" lion is the my fav animal");
    }
}

class bird extends animal{
    String names = " parrot , sparrow";
    void bi(){
        System.out.println("parrot is my fav bird");// multi inheritanceis executed
    }
}

class sea extends bird {
    float marks = 99.23f;
    void mar(){
        System.out.println(" you got 1 st rank");
    }
}

public class Inherit2 {
    public static void main(String[] args) {

        // we create an object 3 rd class
        sea as = new sea();

        /*
        System.out.println(as.name + " my fav animal");
        System.out.println(as.names + "my fav bird");
        System.out.println(as.marks + "i got this rank");*/

        as.mar();
        as.bi();
        as.ani();
    }
}
```

# Output for second type of inheritance

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Inherit2.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Inherit2
you got 1 st rank
parrot is my fav bird
lion is the my fav animal
```

# Third type:

```
// hiarchearicalinheritance type is 3rd type of inheritance
// we create one super class is inherited by two sub class
    class animal{
        String name = " lion";
        void ani(){
            System.out.println(" lion is the my fav animal");
        }
    }

    class bird extends animal{
        String names = " parrot , sparrow";
        void bi(){
            System.out.println("parrot is my fav bird");// multi inheritanceis executed
        }
    }

    class sea extends animal {
        float marks = 99.23f;
        void mar(){
            System.out.println(" you got 1 st rank");
        }
    }

    public class Inherit3 {
        public static void main(String[] args) {
            // create an object for 3rd class to super class
            sea as = new sea();
            /*
            System.out.println(as.name + " my fav animal");
            System.out.println(as.names + "my fav bird");
            System.out.println(as.marks + "i got this rank");*/

            as.mar();
            as.ani();
            as.ani();
        }
    }
```



# output

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Inherit3.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Inherit3
you got 1 st rank
lion is the my fav animal
lion is the my fav animal
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> 
```

# Fourth type :

```
// type : 4
//hybridinheritance of using two types of inheritance are used
//! single and multi inheritance are used
// single inheritance
class fir{
    int a =10, b=90;
    void rif(){
System.out.println(" addition is " + (a+b));
    }
}

class sec extends fir{
    double c = 2.34f,d = 4.56f;
    void ces(){
System.out.println(" th  eresult is " + (c+d));
    }
}

// multi inheritance are starting
class thi extends sec{
    String name ="heamnth , phk";

    void hit(){
System.out.println(" name is" + name);
    }
}

class four extends sec{
    int f =32,g=34;
    void rouf(){
System.out.println(" result is " + (f*g));
    }
}

public class Inherit4 {
    public static void main(String[] args) {
// we can create object for 4 th sub class
        four oj = new four();
        oj.rouf();
        oj.ces();
        oj.ces();
        sec jo = new sec();
        jo.rif();
        jo.ces();
        thi joj = new thi();
        joj.hit();
        joj.ces();

    }
}
```



```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Inherit4.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Inherit4
result is 1088
the result is 6.8999998569488525
the result is 6.8999998569488525
addition is 100
the result is 6.8999998569488525
name ishemanth , phk
the result is 6.8999998569488525
```

# Type 5 :

```
/*
this 5 th type of inheritance is called multiple inheritance
*/

//we create two interfaces called first and second
// in first interface having a initialised variable and abstract method
interface first{
    int a =244;
    void system();
}

// in second interface have an initialised variable and abstract method called part
interface second{
    int y = 300;
    void part();
}

// to inherit the interfaces by using implements keyword
public class MultipleInherit implements first , second {
    // to override the methods
    @Override
    public void system(){
        System.out.println("the multiple inheritance value is" + a);
    }
    @Override
    public void part(){
        System.out.println("the second interface of value is " + y);
    }

    public static void main(String[] args) {
        // we create an object for class
        MultipleInherit mi = new MultipleInherit();
        // we call methods in main method
        mi.system();
        mi.part();

    }
}
```

# Multiple inheritance output

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac MultipleInherit.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java MultipleInherit
the multiple inheritance value is 244
the second interface of value is 300
```